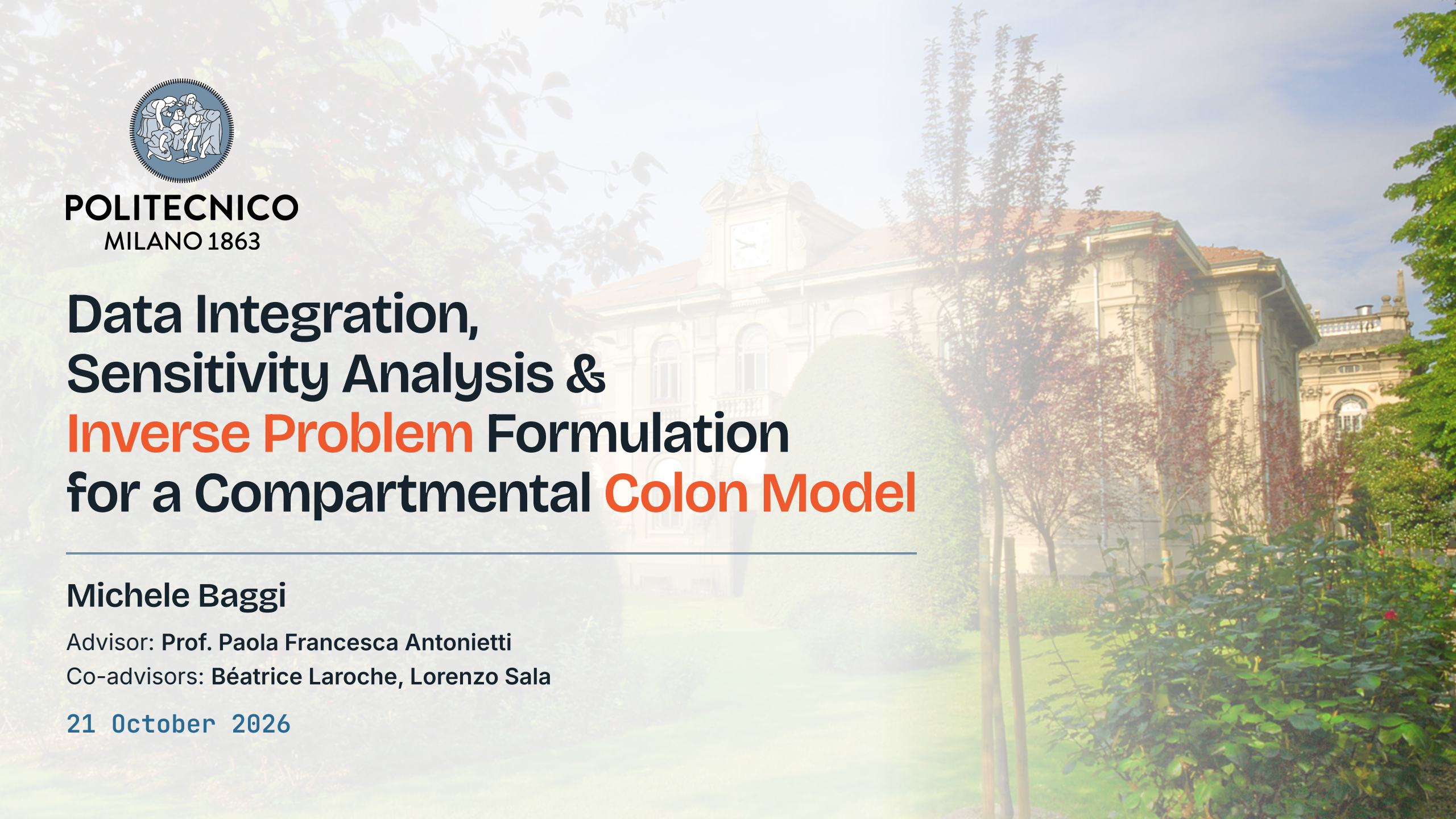


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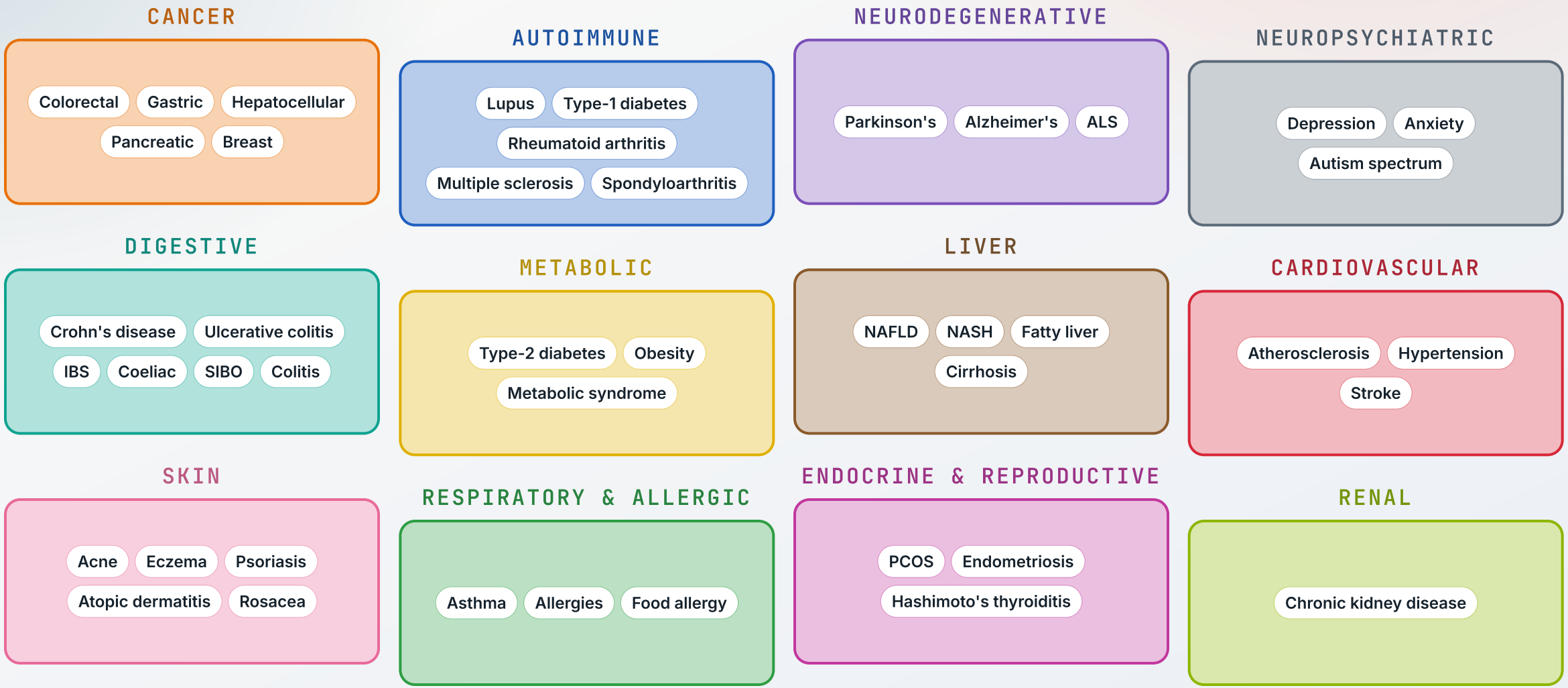
Data Integration, Sensitivity Analysis & Inverse Problem Formulation for a Compartmental Colon Model

Michele Baggi

Advisor: **Prof. Paola Francesca Antonietti**

Co-advisors: **Béatrice Laroche, Lorenzo Sala**

21 October 2026



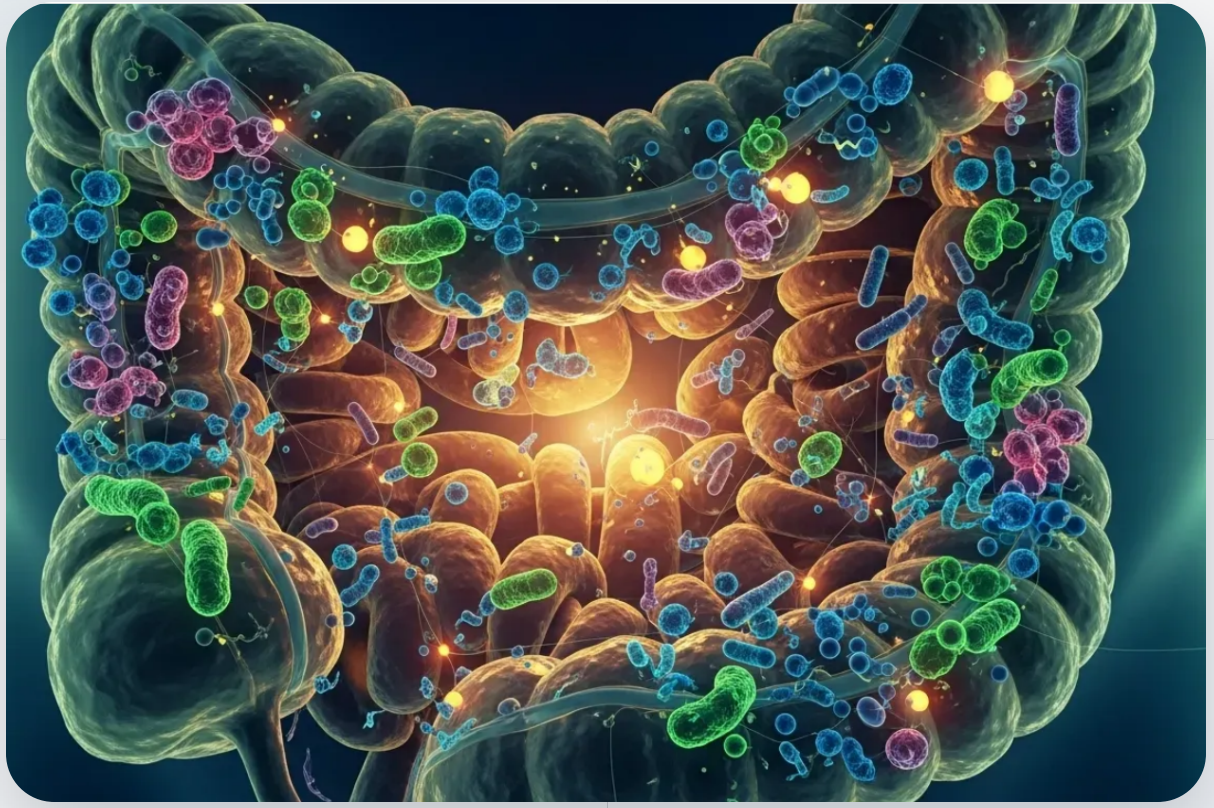
What do they all have in common?

“All disease begins in the gut”

— attributed to Hippocrates, c. 400 BCE

An overstatement, but today we know that a **disrupted gut microbiome** is associated with all of these conditions.

But why is it so important?



~4×10¹³

bacterial cells — about as many as our own

70%

of the immune system is shaped in the gut

90%

of the body's serotonin is produced there

It even affects the way we think.

What is a good microbiome?

Rely on a patient-specific digital twin of the colon that can predict how the gut microbiome will respond to a perturbation (e.g., diet, drugs, probiotics, prebiotics, faecal transplant).



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DIPARTIMENTO DI MATEMATICA

Mathematical Engineering

What I was given

Data

data from **3613 faecal samples** of many patients:

`six source cohorts — different health conditions and nationalities`

Model

a compartmental ODE model of the gut of one **generic patient**

What I did

Make the physical model **patient-specific** using the data, one sample at a time.

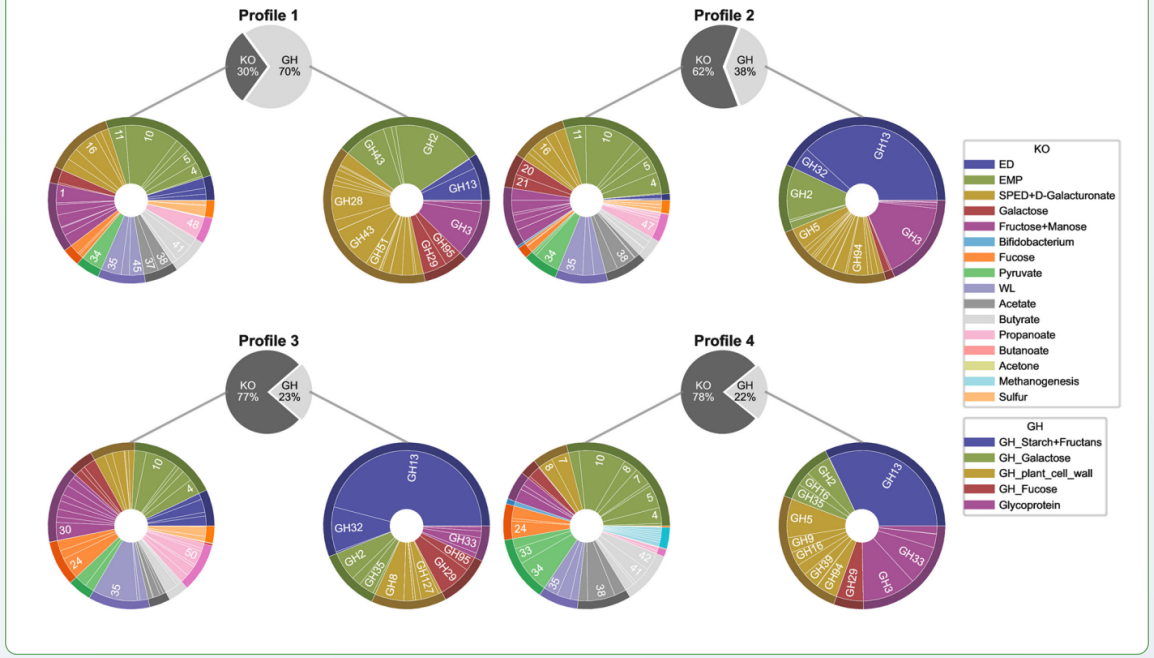
Functions, not species

INPUT	SHOTGUN	MAP	NMF	OUTPUT
1	**9.9M**	**101**	**4**	**$W_1 \ldots W_4$**
sample one time point	gene catalogue	fibre & mucin functional traits	functional profiles	non-negative weights



genes
↳ encode
enzymes
↳ catalyse
reactions
↳ together form a
function

One gene can contribute to one or more functions.

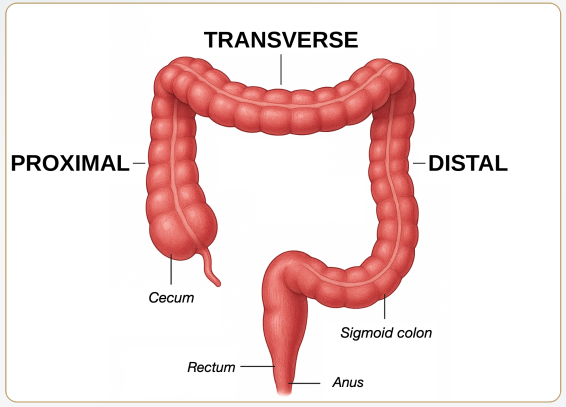


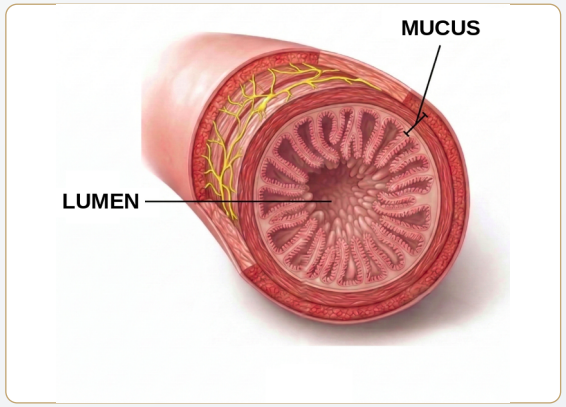
Each sample = $\Sigma\ w_k \cdot \text{profile}_k$, with $w_k \geq 0$ and $\Sigma\ w_k$ **not** constrained (Labarthe et al. 2023).

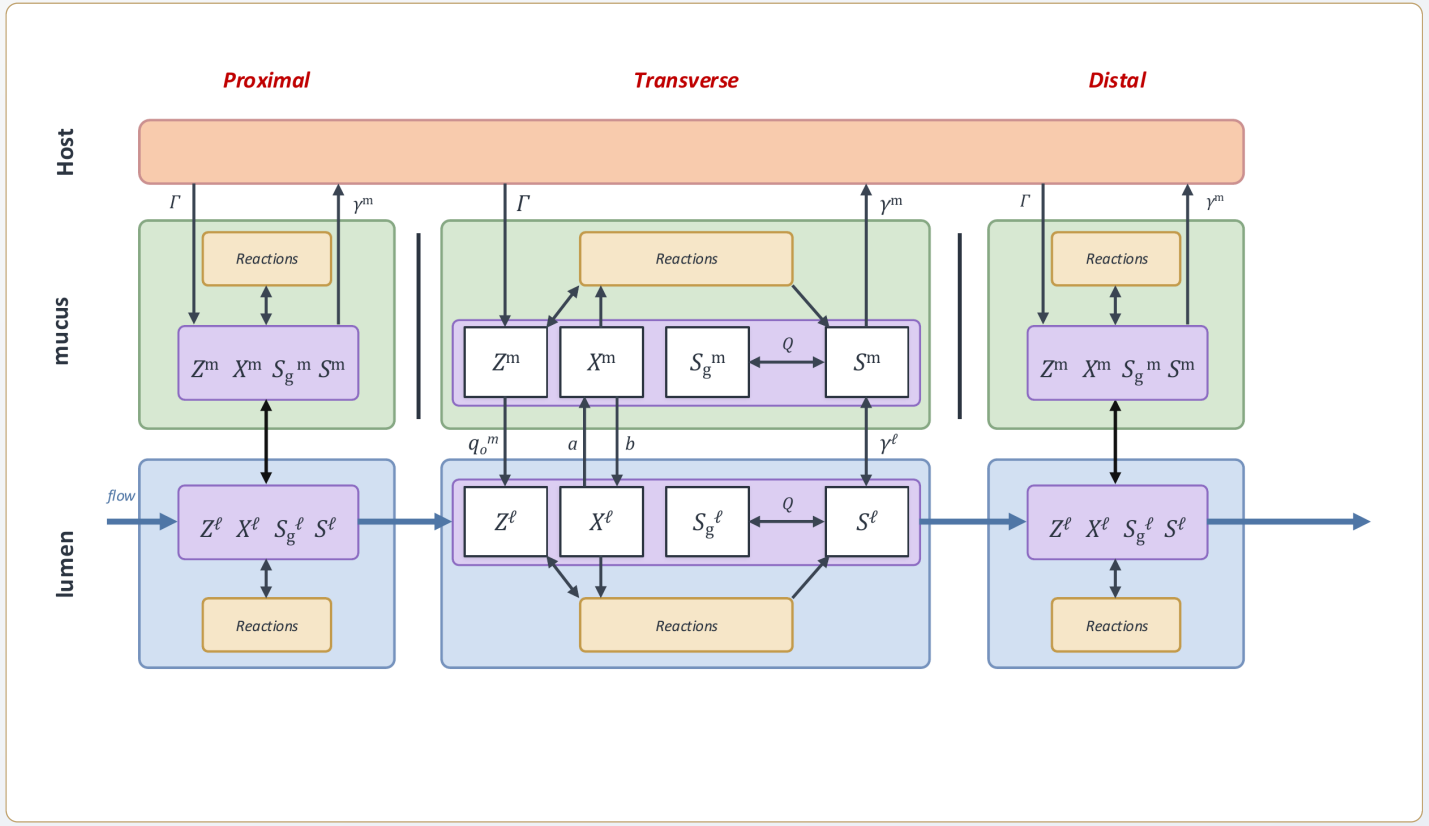
A mechanistic model of a generic patient's colon

3 regions in series, 2 layers (6 compartments)

State variables are uniform in space, but changing in time. **4** bacteria populations are considered.



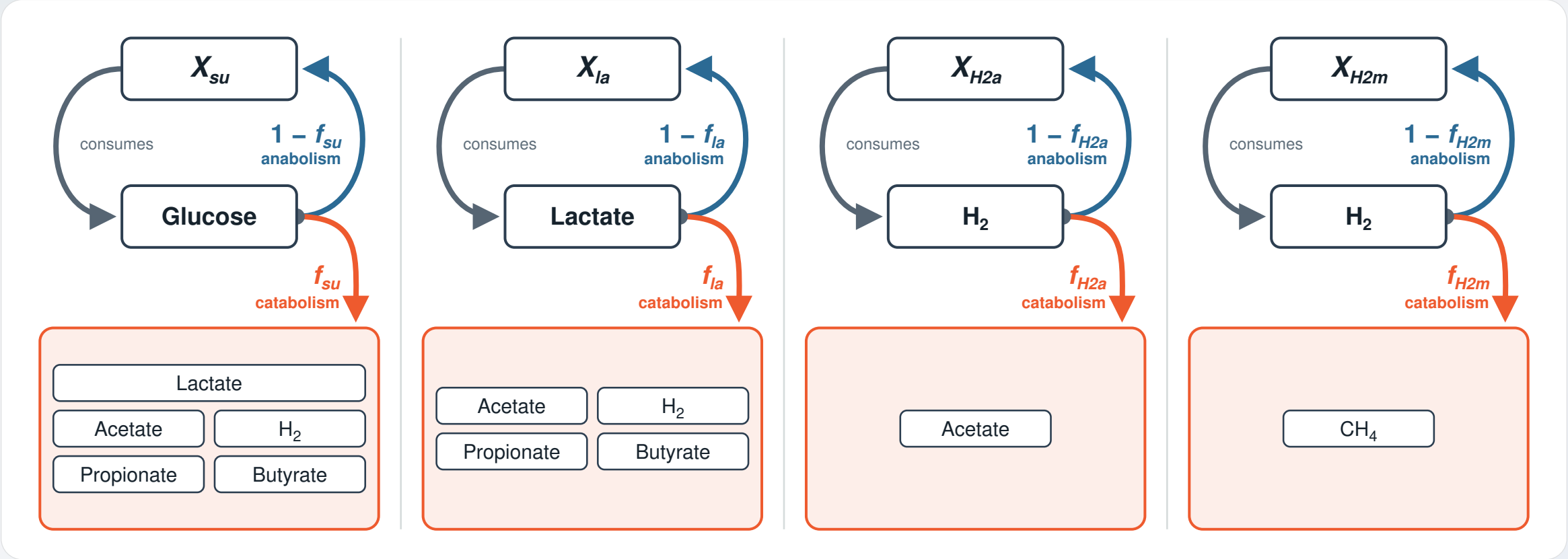




$$\dot{\mathbf{u}} = F(\mathbf{u}, \boldsymbol{\theta}, t),\ \mathbf{u} \in \mathbb{R}^{106}$$ · reset at each excretion (**hybrid**)

Carbohydrate fermentation

Each of the 4 populations consumes one substrate



f_i → catabolism (new metabolites) · $1\text{-}f_i$ → anabolism (new biomass) ⇒ **4 independent parameters**

The inverse problem

Find parameters so the model **reproduces** a real patient's microbial composition.

01
Match
Adapt the model and define a QoI to compare it with data.

02
Sensitivity
Which parameters actually move the output?

03
Surrogate
Replace the ODE solver with a fast neural emulator.

04
MCMC
Bayesian estimation of the relevant parameters.

Match the physical model with data

Generalist populations

x_1: glucose

x_2: α_1 glucose + α_2 lactate + α_3 acetogenesis

x_3: β_1 lactate + β_2 acetogenesis

x_4: methanogenesis

`α = (0.5, 0.3, 0.2) · β = (0.8, 0.2)`

→

Match proportions

Profile weights are not concentrations: only compositions can be compared.

$$\mathbf{y} \triangleq \frac{\mathbf{x}_D^l(t_e)}{\|\mathbf{x}_D^l(t_e)\|_1} \approx \frac{\mathbf{w}}{\|\mathbf{w}\|_1}$$

Aitchison distance

distance between compositions

$$d_a(\mathbf{x},\mathbf{y}) = \left\|\mathrm{clr}(\mathbf{x}) - \mathrm{clr}(\mathbf{y})\right\|_2, \qquad \mathrm{clr}(\mathbf{x})_i = \ln \frac{x_i}{g(\mathbf{x})}$$

g = geometric mean · compares log-ratios, so only relative abundances count

Sensitivity Analysis

Which parameters actually move the relative abundances

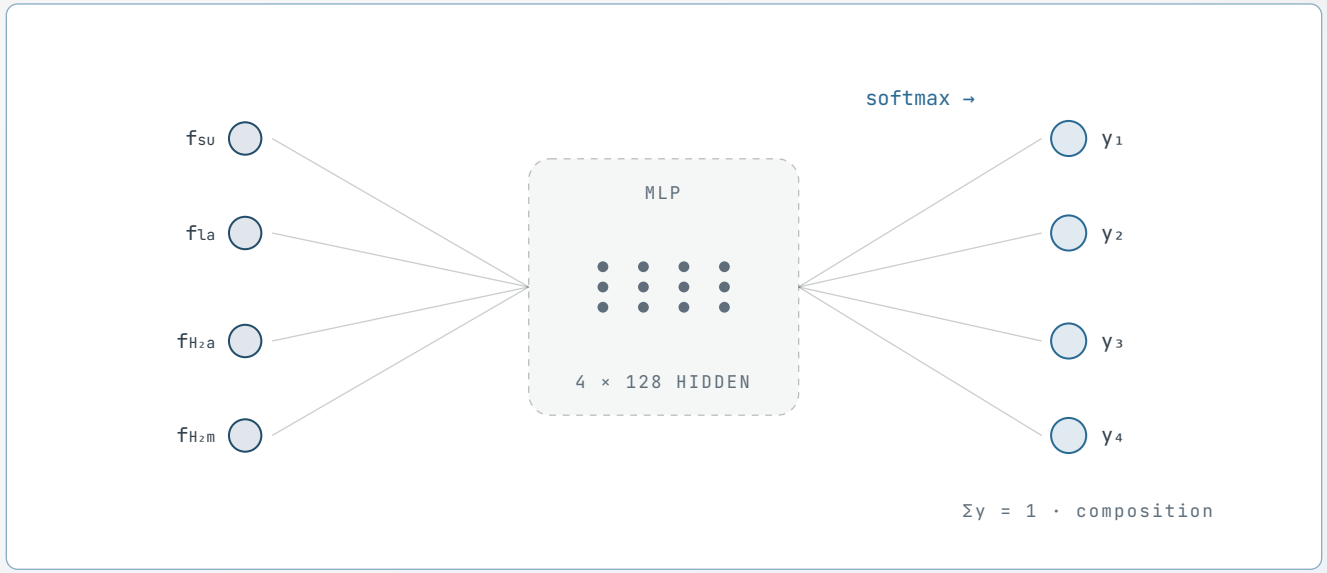
60 parameters → biological reasoning → Morris screening → Sobol indices → **4** retained

$$\vec{f} = \left(f_{\mathrm{su}},\ f_{\mathrm{la}},\ f_{\mathrm{H_2a}},\ f_{\mathrm{H_2m}}\right)$$

the four catabolic fractions: the parameters inferred from now on

A neural surrogate

One run of the full model takes several seconds — too slow inside MCMC. An MLP $\hat{G}$ replaces it, trained on $2^{15} = 32\,768$ scrambled-Sobol' vectors $\vec{f}$ run through the ODE model.



$$\mathcal{L}(\phi) = \frac{1}{N} \sum_{\vec{f} \in \mathcal{D}} d_a^2\left(\widehat{G}_\phi(\vec{f}),\, G(\vec{f})\right)$$

G = ODE model (target composition) · $\hat{G}_\psi$ = MLP with weights ψ · $\mathcal{D}$ = training set of N vectors · Test set: median d_a 0.012, 95th percentile 0.060, $R^2 > 0.998$ in clr coordinates.

Bayesian formulation & MCMC

The parameters are random variables: the answer is a **posterior distribution** over $\vec{f}$, given the sample.

PROPOSE
$\vec{f}*$
candidate parameters

→

SURROGATE
$\widehat{G}(\vec{f}*)$
predicted composition

→

COMPARE
d_a
distance to the sample $\mathbf{z}$

→

ACCEPT / REJECT
$\vec{f}^{(n+1)}$
next state of the chain

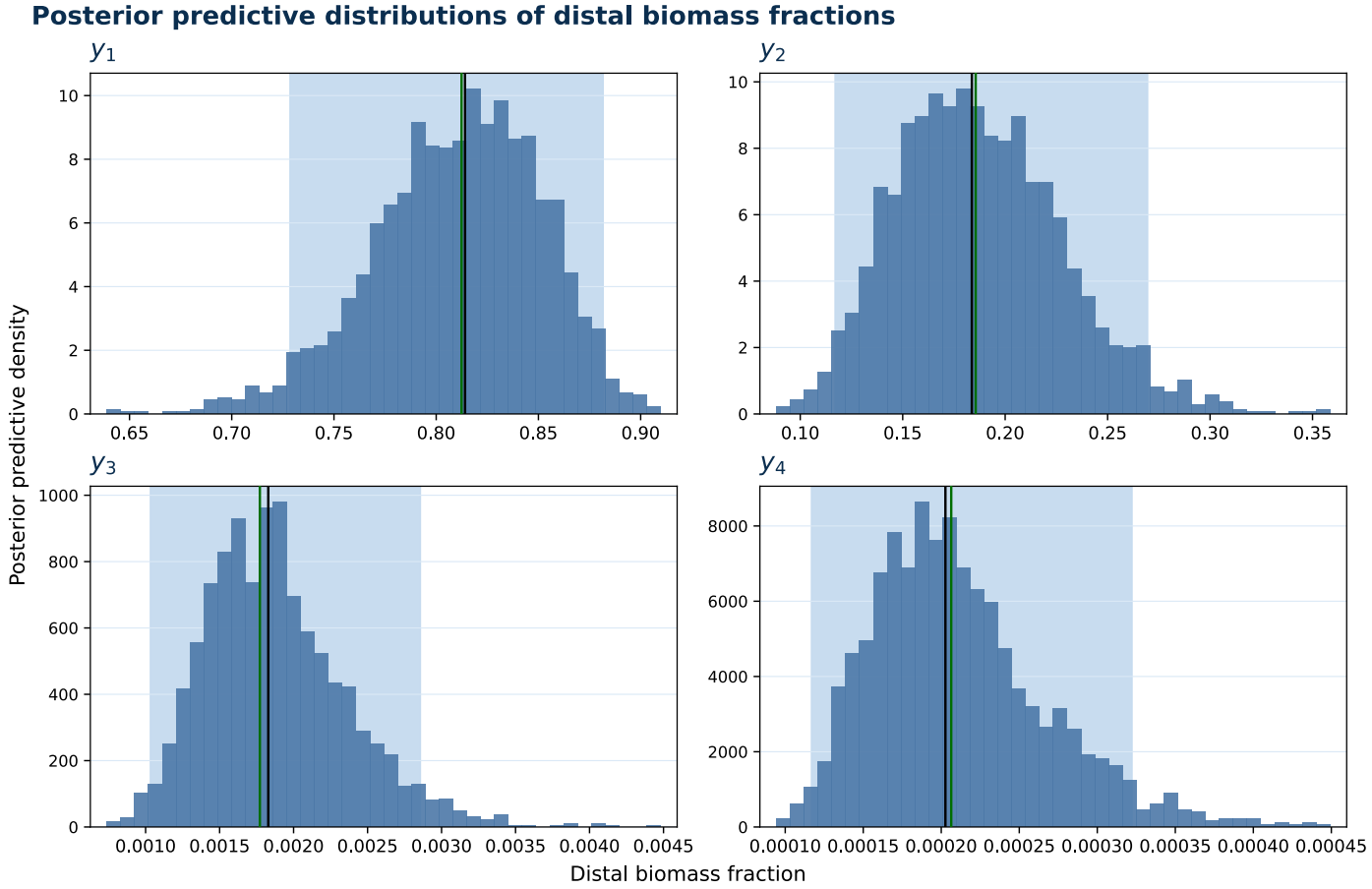
$$\underbrace{\log \pi(\mathbf{z} \mid \vec{f})}_{\text{likelihood}} \propto -\frac{1}{2\sigma^2}\, d_a^2\big(\widehat{G}(\vec{f}), \mathbf{z}\big), \qquad \sigma = 0.2$$

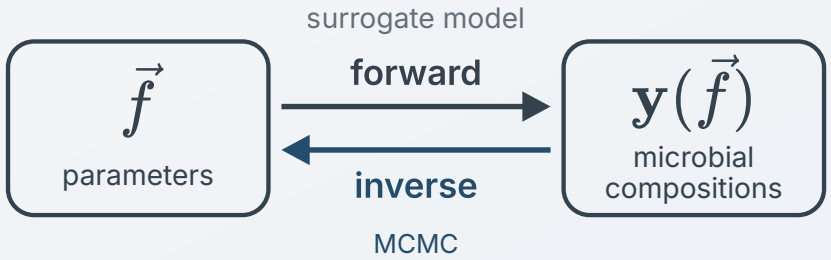
the closer $\widehat{G}(\vec{f}*)$ is to $\mathbf{z}$, the more likely $\vec{f}*$ is accepted

Repeating this step generates a chain $\vec{f}^{(1)}, \vec{f}^{(2)}, \ldots$ whose states are **distributed as the posterior**.

Sampler used: DEMetropolis(Z), a Metropolis variant with adaptive proposals, in PyMC (appendix)

One sample — the observation is recovered





Medoid of the 3613 samples (smallest mean Aitchison distance to all others).

6 chains, from different starting points agree: $\hat{R} \leq 1.002$

Shaded: 95 % interval · black: predictive median · green: observation

The whole cohort

3613/3613

samples inverted, none failed
2 h 46 min on 64 cores

0.142

median distance **model − data**
full ODE model at the posterior median, 100 random samples

3.24

mean distance **between subjects**
(to the cohort centre)

The model reproduces each sample **an order of magnitude more closely** than subjects differ from each other.

What this work delivers

Done

- model **re-implemented** and adapted: its four groups match the four functional profiles
- **global sensitivity analysis**: 60 parameters → the 4 catabolic fractions
- Bayesian inversion with a neural surrogate on all **3613 samples**, fit confirmed with the full model

Next

- new measurements — **SCFAs, total biomass, repeated samples, diet** — as extra constraints
- a **richer representation** of the metagenome than four profiles
- the same workflow on **richer models**

Each sample becomes a model that can be **simulated**
→ a first step towards a per-patient gut digital twin

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Thank you

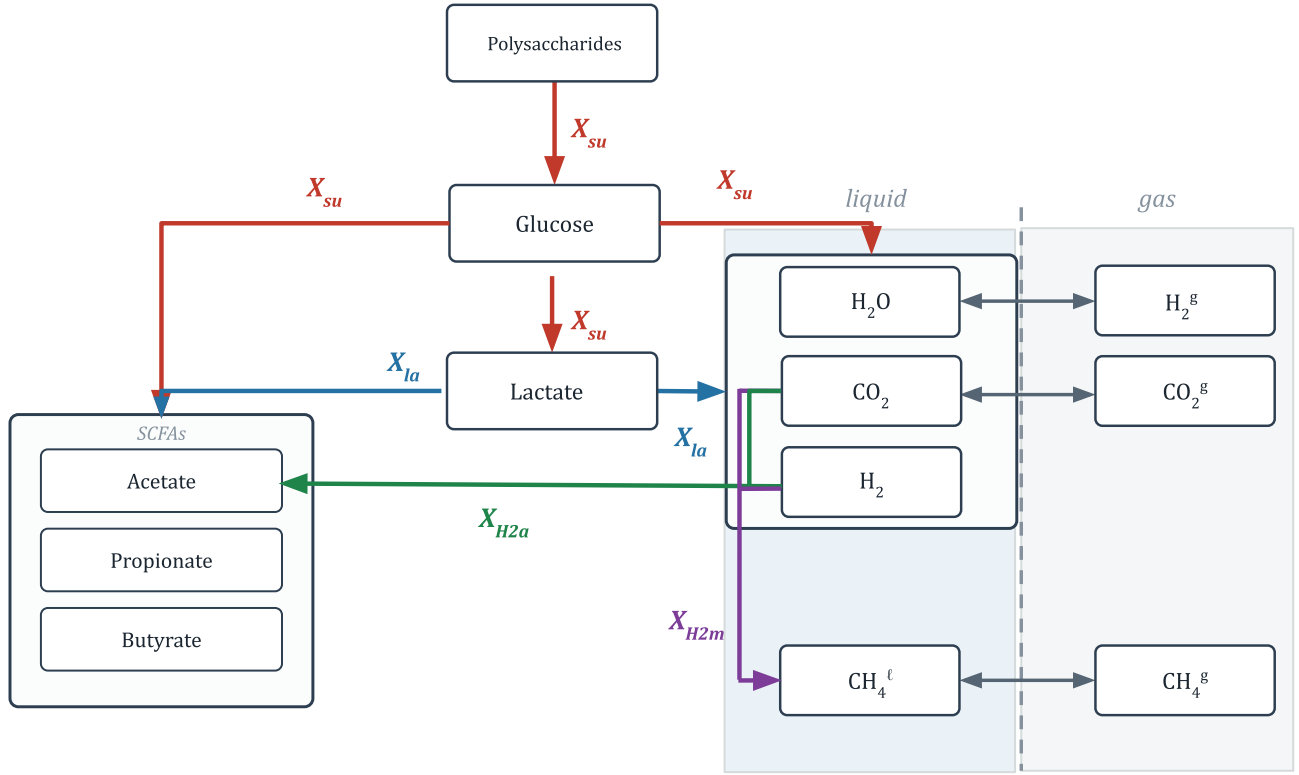
Questions welcome

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- Muñoz-Tamayo et al. — Mathematical modelling of carbohydrate degradation by human colonic microbiota, *J. Theor. Biol.* 266 (2010).
- Raguideau et al. — Inferring aggregated functional traits from metagenomic data using constrained NMF, *PLoS Comput. Biol.* 12 (2016).
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Appendix · Carbohydrate fermentation

Bacteria convert substrates into metabolites



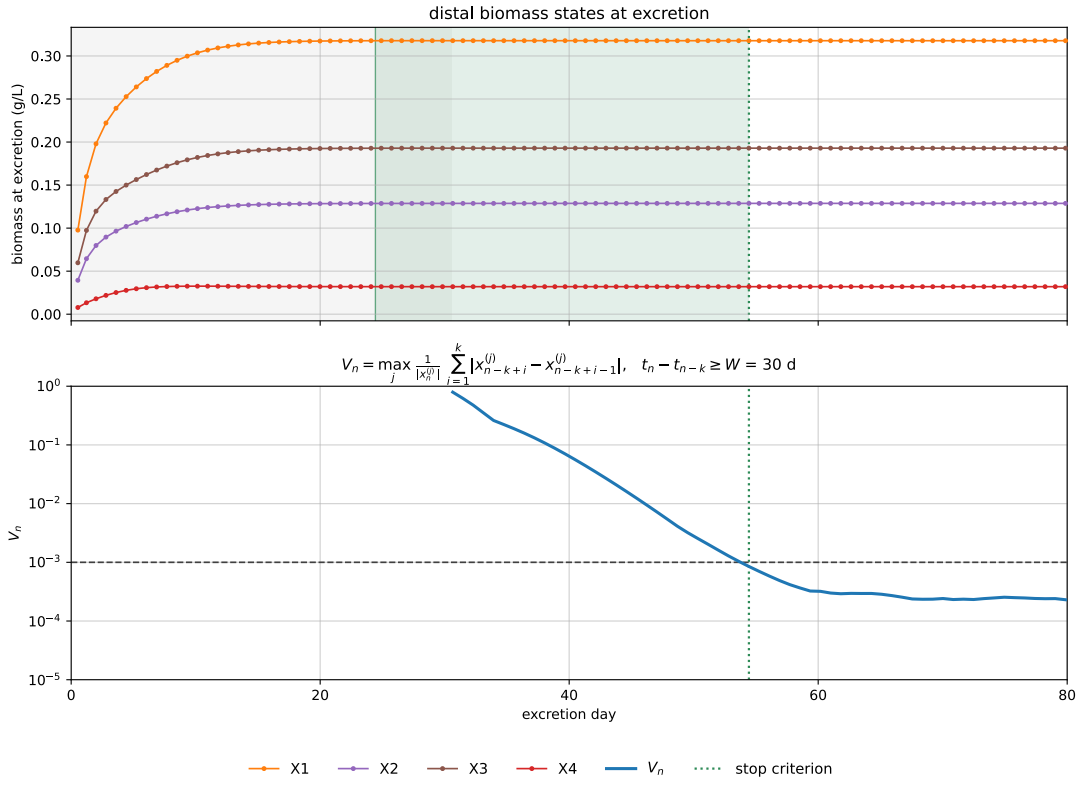
f → catabolism
1–f → anabolism (growth)

η — split of the catabolic products

k_{dec} — microbial decay rate

Appendix · Convergence criterion

For each distal biomass, sum its absolute changes at excretion over a 30-day window and divide by its current value; stop when the maximum over the four groups is below 10^{-3}.



Appendix · Why Aitchison distance?

Model and data are **compositions** on the simplex (Σ = 1). Euclidean distance on raw fractions is misleading — it ignores the constant-sum constraint and isn't scale-invariant.

Centred log-ratio (CLR)

$$\mathrm{clr}(x)_i = \ln \frac{x_i}{g(x)}$$

$$g(x) = \left(\prod_{j=1}^{D} x_j \right)^{1/D}$$

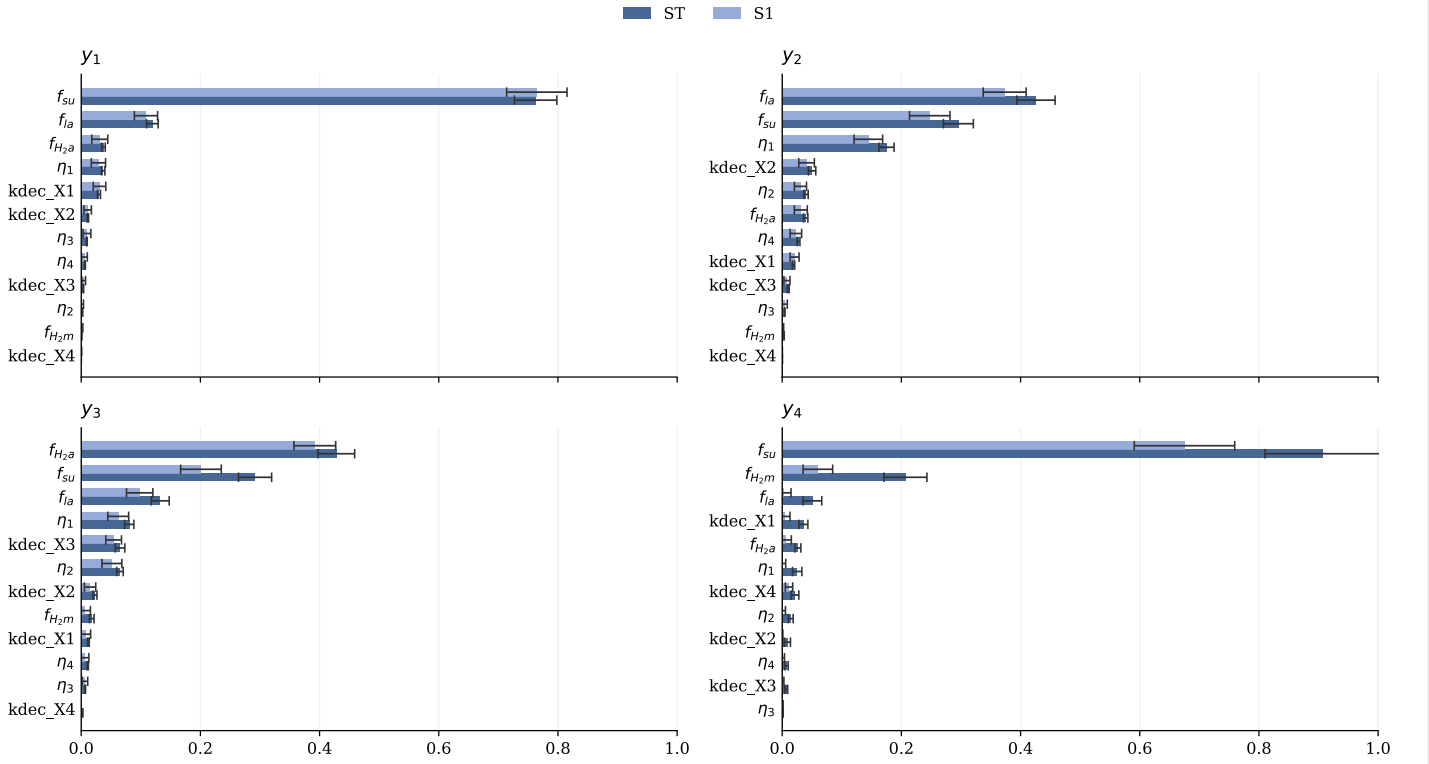
g(x) = geometric mean · maps the simplex into ordinary real space (D = 4 here).

Aitchison distance

$$d_a(x, y) = \left\| \mathrm{clr}(x) - \mathrm{clr}(y) \right\|_2 = \sqrt{\sum_i \left(\mathrm{clr}(x)_i - \mathrm{clr}(y)_i \right)^2}$$

Euclidean distance *after* CLR — scale-invariant & subcompositionally coherent.

Appendix · Sobol' indices — the 12 factors retained by Morris



Setup

Factors: 4 f · 4 η (glucose split) · 4 k_{dec}
Ranges as in Morris: factor ×/÷ 2 (f: log-odds ±log 2; k_{dec}: ±1 decade)
Saltelli et al. (2010), second order: N(2k+2) = **53 248** runs, N = 2048
Error bars: bootstrap confidence intervals

Σ S1

	Y_1	Y_2	Y_3	Y_4
f block	0.91	0.65	0.70	0.74
all 12	1.00	0.92	0.90	0.75

Additive on y_1–y_3. On y_4 f_{su}: S1 0.68 → ST 0.91; largest pair (f_{su}, f_{H_2m}), S2 = 0.08.

k_{dec}: ST ≤ 0.07 despite a 10× wider range · η: max ST 0.18 (η_1 on y_2), else ≤ 0.09

S1 = Var(E[Y|θᵢ]) / Var(Y): θᵢ alone · ST = E[Var(Y|θ₋ᵢ)] / Var(Y): with all interactions · y° = (0.471, 0.253, 0.252, 0.024)

Appendix · Surrogate — architecture, training, accuracy



hidden layers, tanh

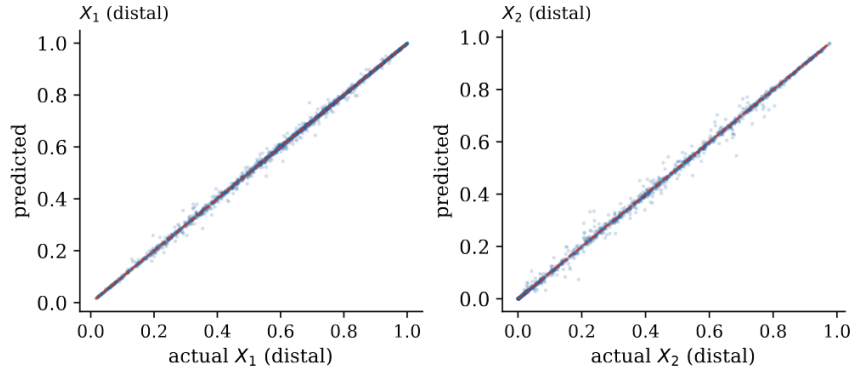
TRAINING

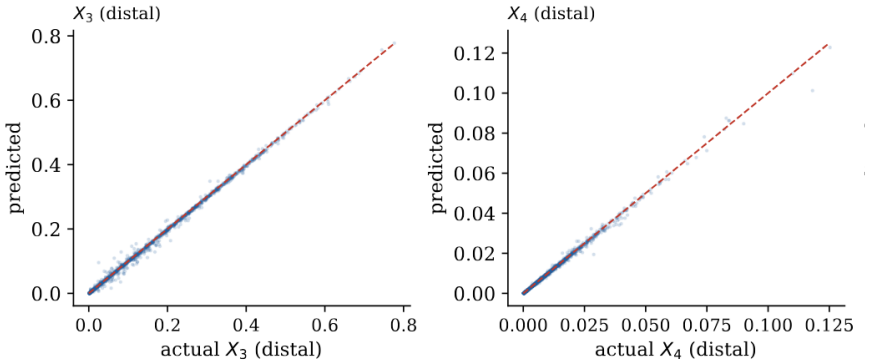
design	2^{15} = 32 768 Sobol' points
split	80 / 10 / 10 %
loss	squared Aitchison distance
optimiser	Adam, batch 1024, 2000 epochs
learning rate	cosine, $1.5 \cdot 10^{-3} \rightarrow 10^{-5}$
best val. loss	epoch 1485 · ≈ 9 min
weights	50 692

TEST SET

	Y_1	Y_2	Y_3	Y_4
R^2 (clr)	0.99980	0.99861	0.99986	0.99914
log-ratio RMSE	0.0079	0.0808	0.0325	0.0611

AITCHISON	MEAN	MEDIAN	P95
	0.0254	0.0121	0.0601





test set · log axes · dashed = identity

Appendix · Bayesian formulation

$$\underbrace{\pi(\vec{f} \mid \mathbf{z})}_{\text{posterior}} = \frac{\overbrace{\pi(\mathbf{z} \mid \vec{f})}^{\text{likelihood}}\ \overbrace{\pi_{\text{prior}}(\vec{f})}^{\text{prior}}}{\int \pi(\mathbf{z} \mid \vec{f})\, \pi_{\text{prior}}(\vec{f})\, d\vec{f}}$$

Likelihood

$$\log \pi(\mathbf{z} \mid \vec{f}) = -\frac{1}{2\sigma^2}\, d_a^2\big(\widehat{G}(\vec{f}), \mathbf{z}\big) + \text{const}, \qquad \sigma = 0.2$$

Ĝ = neural surrogate · d_a = Aitchison distance · σ = tolerance on the model–data misfit (isotropic Gaussian error in clr coordinates)

Prior

$$\vec{f} \sim \mathcal{N}_T\big(\boldsymbol{\mu},\, 0.25^2 I;\, [0.01, 0.99]^4\big)$$

μ = (0.90, 0.80, 0.57, 0.38): reference values of Muñoz-Tamayo et al. (2010) · boundaries excluded to avoid degenerate cases

The normalising integral cannot be computed → the posterior is **sampled** with MCMC (next slide).

Appendix · Metropolis step & DEMetropolis(Z)

Metropolis step

$$\vec{f}^{\,*} = \vec{f}^{\,(n)} + \boldsymbol{\xi}, \qquad \boldsymbol{\xi} \text{ symmetric}$$

$$\alpha = \min\left(1,\ \frac{\pi(\vec{f}^{\,*} \mid \mathbf{z})}{\pi(\vec{f}^{\,(n)} \mid \mathbf{z})}\right), \qquad \vec{f}^{\,(n+1)} = \begin{cases} \vec{f}^{\,*} & \text{w.p. } \alpha \\ \vec{f}^{\,(n)} & \text{otherwise} \end{cases}$$

Only a **ratio** of posteriors: the normalising constant cancels.

DEMetropolis(Z) proposal

$$\vec{f}^{\,*} = \vec{f}^{\,(n)} + \lambda\left(\vec{f}^{\,(r_1)} - \vec{f}^{\,(r_2)}\right) + \boldsymbol{\varepsilon}$$

$\vec{f}^{\,(r_1)}$, $\vec{f}^{\,(r_2)}$ drawn from past states (ε = small noise): the jumps follow the shape of the posterior, no manual tuning. Same acceptance rule.

ter Braak & Vrugt (2008)

MEDOID

6 chains

$5{\cdot}10^4$ tuning + 10^5 draws each · $\hat{R} \leq 1.002$ · ESS > 7800

WHOLE COHORT · 3613 SAMPLES

4 chains

8000 tuning + 8000 draws each · 2 h 46 min on 64 cores